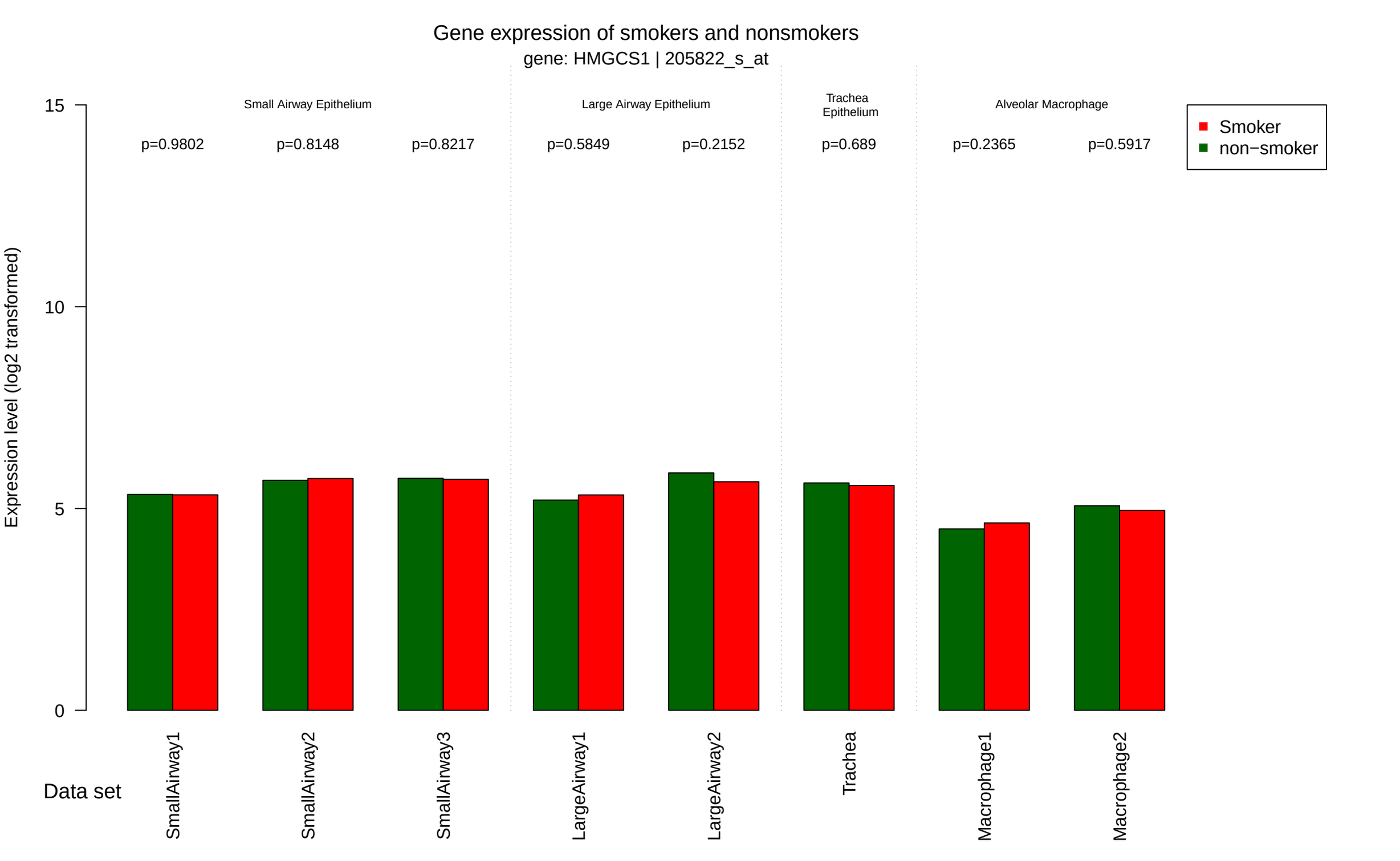
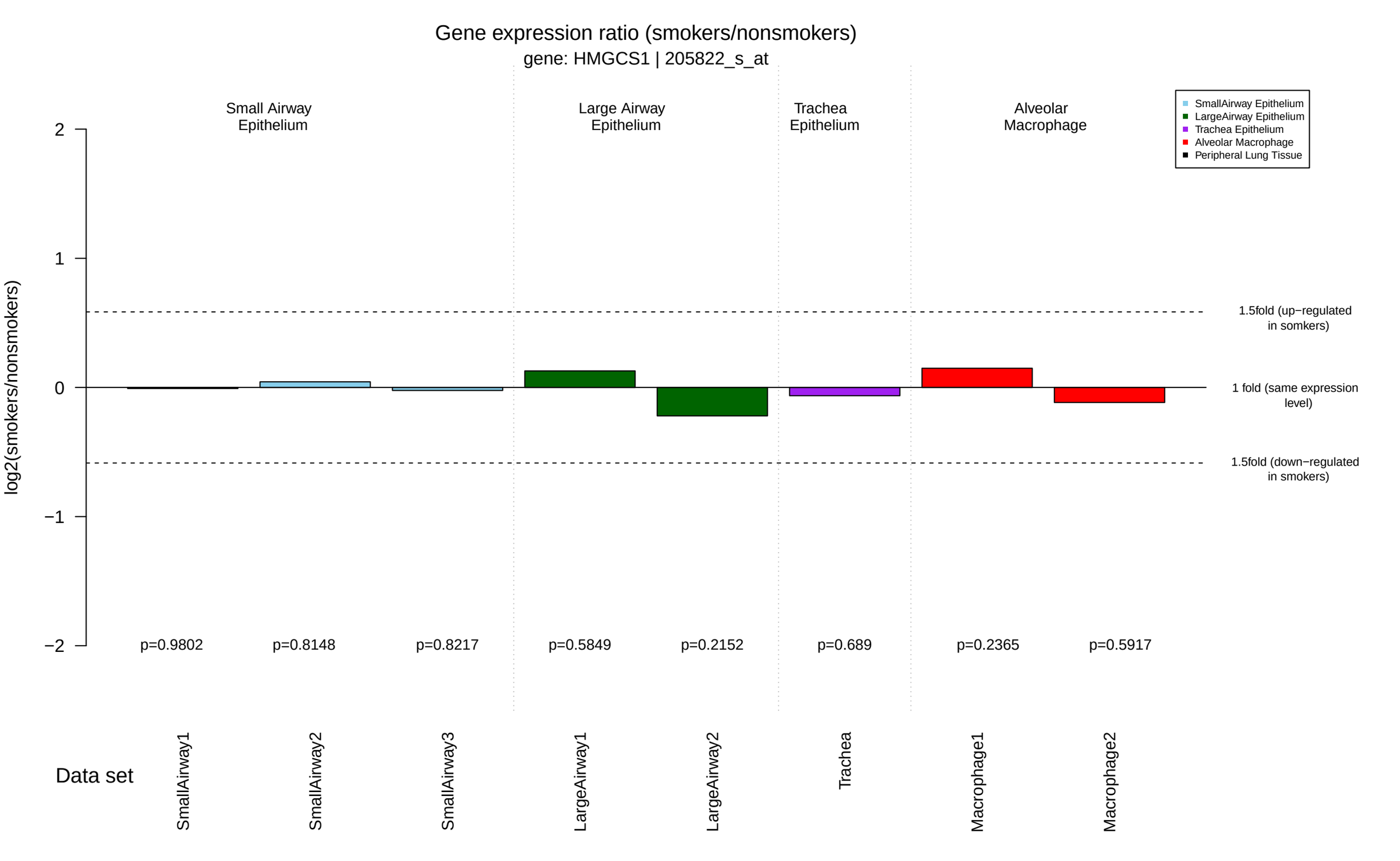
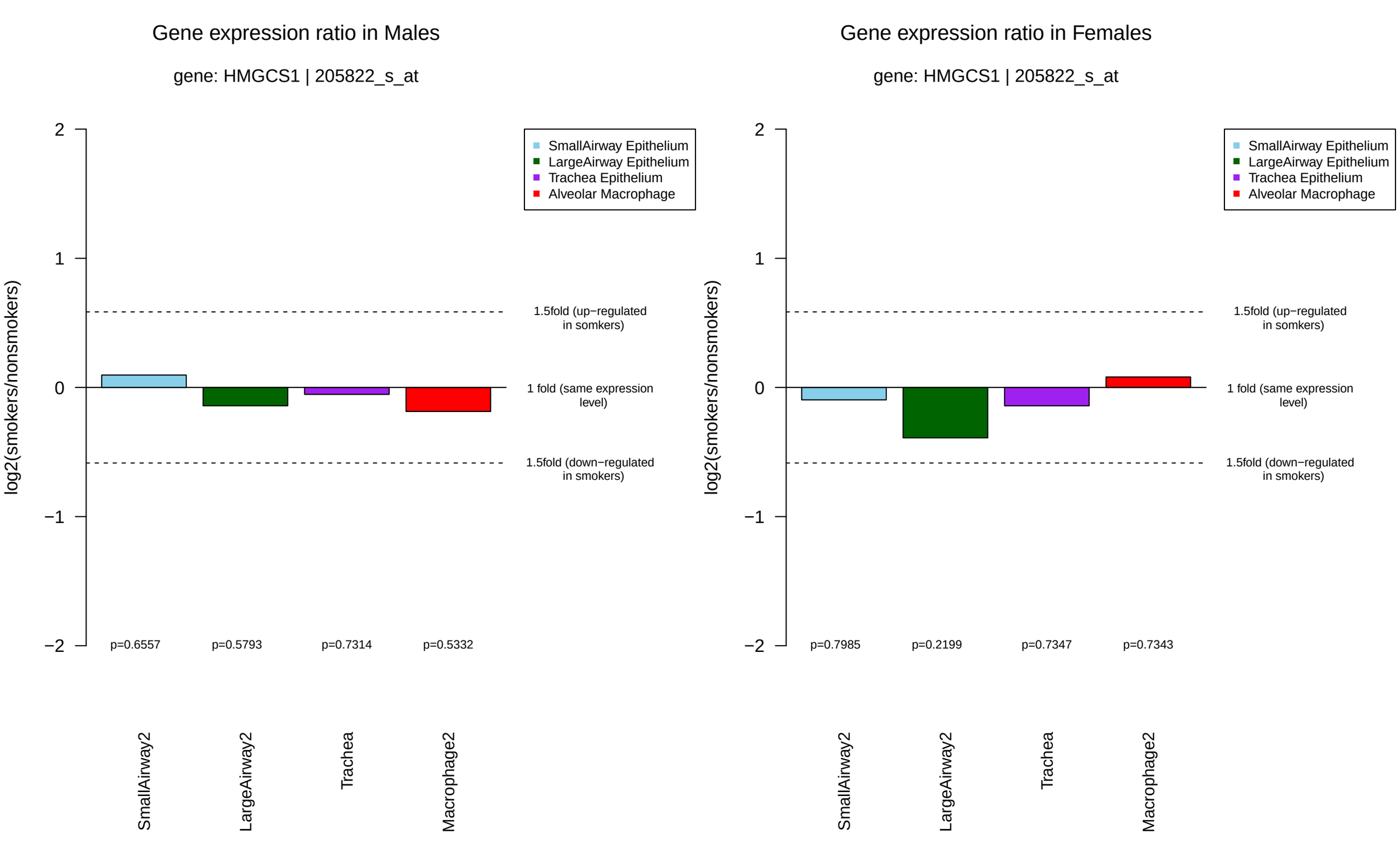
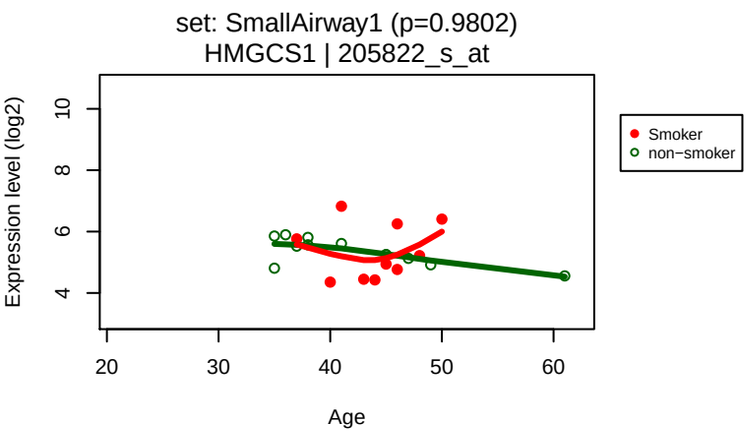
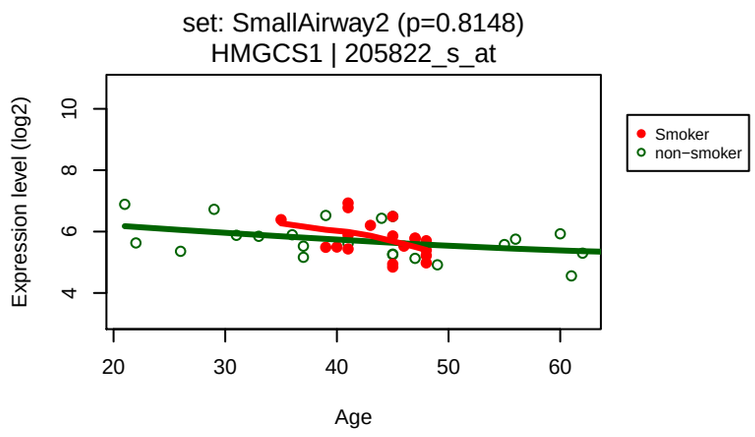
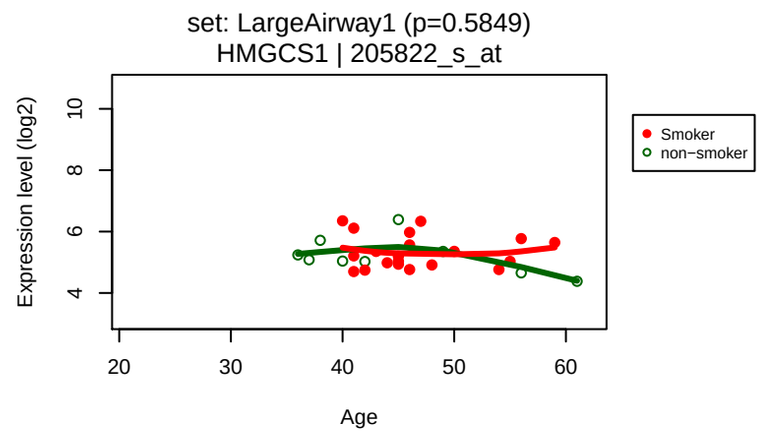

Gene expression of smokers and nonsmokers
gene: HMGCS1 | 205822_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9802
p=0.8148
p=0.8217
p=0.5849
p=0.2152
p=0.689
p=0.2365
p=0.5917
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

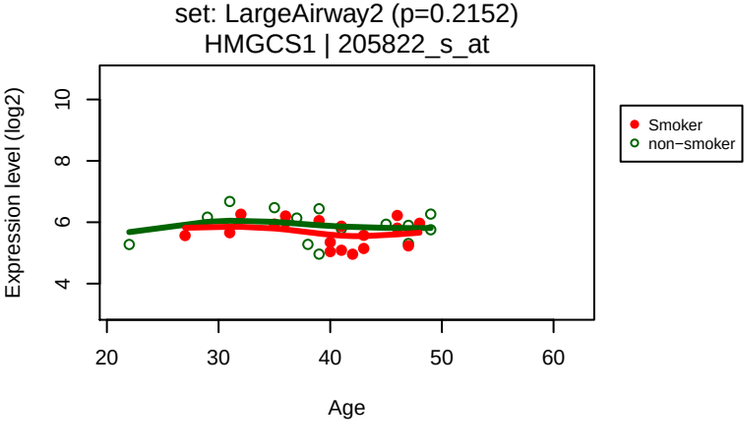

Gene expression ratio (smokers/nonsmokers)
gene: HMGCS1 | 205822_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9802
p=0.8148
p=0.8217
p=0.5849
p=0.2152
p=0.689
p=0.2365
p=0.5917
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

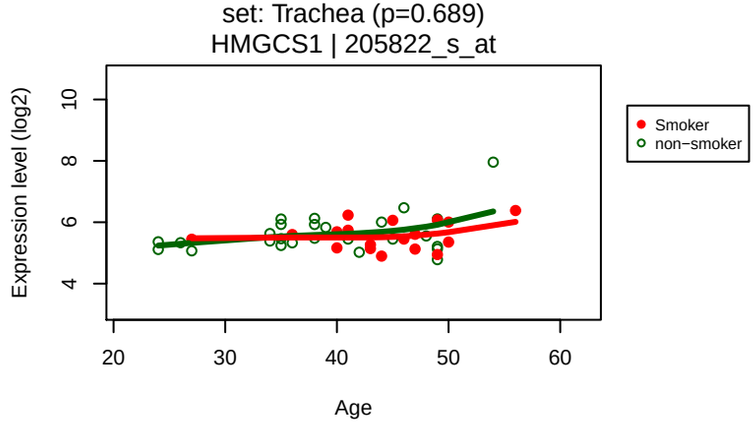

Gene expression ratio in Males
gene: HMGCS1 | 205822_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6557
p=0.5793
p=0.7314
p=0.5332
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HMGCS1 | 205822_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7985
p=0.2199
p=0.7347
p=0.7343
SmallAirway2
LargeAirway2
Trachea
Macrophage2

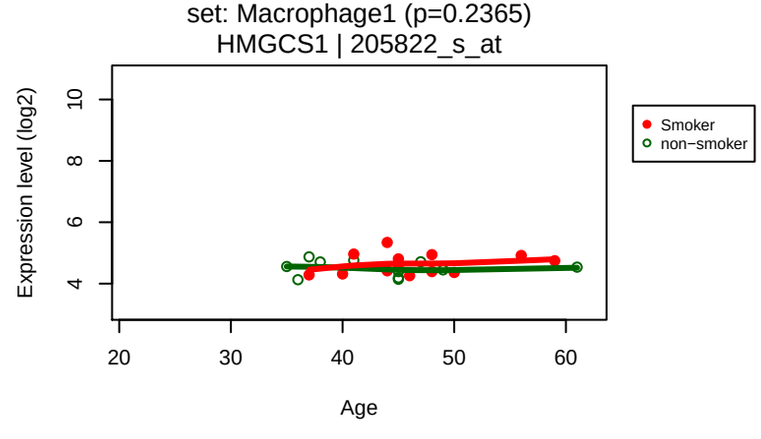

set: SmallAirway1 (p=0.9802)
HMGCS1 | 205822_s_at
Expression level (log2)
Age
Smoker
non−smoker

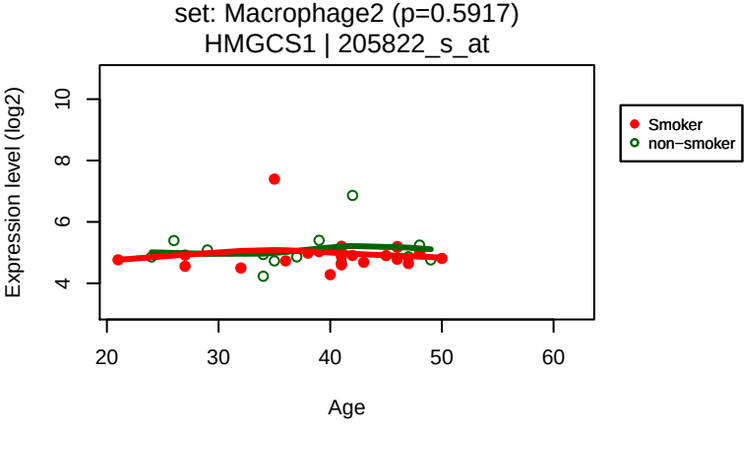

set: SmallAirway2 (p=0.8148)
HMGCS1 | 205822_s_at
Expression level (log2)
Age
Smoker
non−smoker

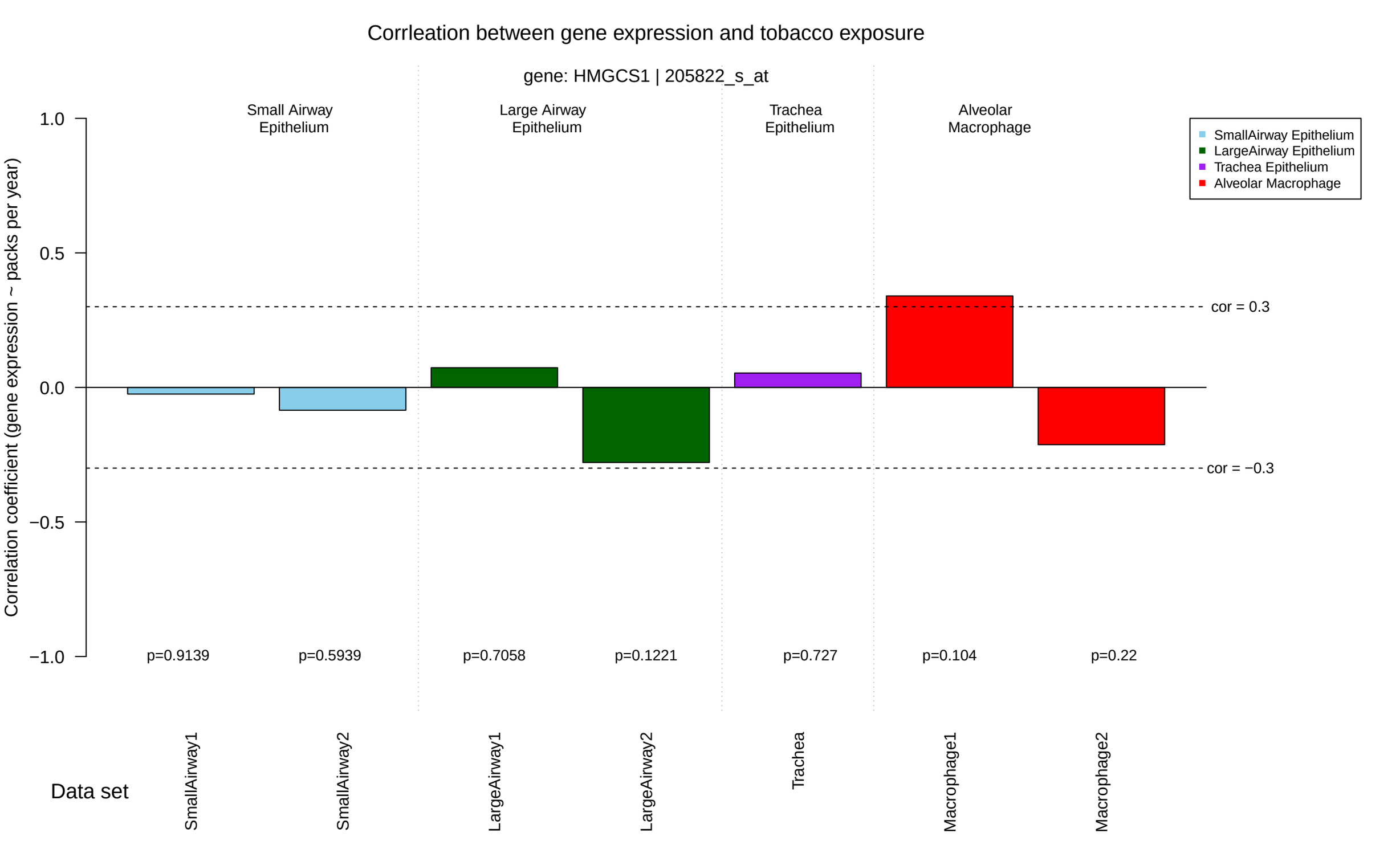

set: LargeAirway1 (p=0.5849)
HMGCS1 | 205822_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2152)
HMGCS1 | 205822_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.689)
HMGCS1 | 205822_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2365)
HMGCS1 | 205822_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5917)
HMGCS1 | 205822_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HMGCS1 | 205822_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9139
p=0.5939
p=0.7058
p=0.1221
p=0.727
p=0.104
p=0.22
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2